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Network Types

Each network is structured differently and can be set up individually. For this reason, so-called **types** and **topologies** have been developed that can be used to categorize these networks. When reading about all the types of networks, it can be a bit of information overload as some network types include the geographical range. We rarely hear some of the terminologies in practice, so this section will be broken up into **Common Terms** and **Book Terms**. Book terms are good to know, as there has been a single documented case of an email server failing to deliver emails longer than 500 miles but don't be expected to be able to recite them on demand unless you are studying for a networking exam.

Common Terminology

Network Type	Definition
Wide Area Network (WAN)	Internet
Local Area Network (LAN)	Internal Networks (Ex: Home or Office)

Network Type	Definition
Wireless Local Area Network (WLAN)	Internal Networks accessible over Wi-Fi
Virtual Private Network (VPN)	Connects multiple network sites to one LAN

WAN

The WAN (Wide Area Network) is commonly referred to as [The Internet](#). When dealing with networking equipment, we'll often have a WAN Address and LAN Address. The WAN one is the address that is generally accessed by the Internet. That being said, it is not exclusive to the Internet; a WAN is just a large number of LANs joined together. Many large companies or government agencies will have an "Internal WAN" (also called Intranet, Airgap Network, etc.). Generally speaking, the primary way we identify if the network is a WAN is to use a WAN Specific routing protocol such as BGP and if the IP Schema in use is not within RFC 1918 (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16).

LAN / WLAN

LANs (Local Area Network) and WLANs (Wireless Local Area Network) will typically assign IP Addresses designated for local use (RFC 1918, 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16). In some cases, like some colleges or hotels, you may be assigned a routable (internet) IP Address from joining their LAN, but that is much less common. There's nothing different between a LAN or

WLAN, other than WLAN's introduce the ability to transmit data without cables. It is mainly a security designation.

VPN

There are three main types **Virtual Private Networks** (**VPN**), but all three have the same goal of making the user feel as if they were plugged into a different network.

Site-To-Site VPN

Both the client and server are Network Devices, typically either **Routers** or **Firewalls** , and share entire network ranges. This is most commonly used to join company networks together over the Internet, allowing multiple locations to communicate over the Internet as if they were local.

Remote Access VPN

This involves the client's computer creating a virtual interface that behaves as if it is on a client's network. Hack The Box utilizes **OpenVPN** , which makes a TUN Adapter letting us access the labs. When analyzing these VPNs, an important piece to consider is the routing table that is created when joining the VPN. If the VPN only creates routes for specific networks (ex: 10.10.10.0/24), this is called a **Split-Tunnel VPN** , meaning the Internet connection is not going out of the VPN. This is great for Hack The Box because it provides access to the Lab without the privacy concern of monitoring your internet connection. However, for a company, **split-tunnel** VPN's are typically not ideal because if the machine is infected with malware, network-based detection methods will most likely not work as that traffic goes out the Internet.

SSL VPN

This is essentially a VPN that is done within our web browser and is becoming increasingly common as web browsers are becoming capable of doing anything. Typically these will stream applications or entire desktop sessions to your web browser. A great example of this would be the HackTheBox Pwnbox.

Book Terms

Network Type	Definition
Global Area Network (GAN)	Global network (the Internet)
Metropolitan Area Network (MAN)	Regional network (multiple LANs)
Wireless Personal Area Network (WPAN)	Personal network (Bluetooth)

GAN

A worldwide network such as the **Internet** is known as a **Global Area Network (GAN)**. However, the Internet is not the only computer network of this kind. Internationally active companies also maintain isolated networks that span several **WAN**s and connect company computers worldwide. **GAN**s use the glass fibers infrastructure of wide-area networks and interconnect them by international undersea cables or satellite transmission.

MAN

Metropolitan Area Network (MAN) is a broadband telecommunications network that connects several **LAN**s in geographical proximity. As a rule, these are individual branches of a

company connected to a **MAN** via leased lines. High-performance routers and high-performance connections based on glass fibers are used, which enable a significantly higher data throughput than the Internet. The transmission speed between two remote nodes is comparable to communication within a **LAN**.

Internationally operating network operators provide the infrastructure for **MAN**s. Cities wired as **Metropolitan Area Networks** can be integrated supra-regionally in **Wide Area Networks** (**WAN**) and internationally in **Global Area Networks** (**GAN**).

PAN / WPAN

Modern end devices such as smartphones, tablets, laptops, or desktop computers can be connected ad hoc to form a network to enable data exchange. This can be done by cable in the form of a **Personal Area Network** (**PAN**).

The wireless variant **Wireless Personal Area Network** (**WPAN**) is based on Bluetooth or Wireless USB technologies. A **wireless personal area network** that is established via Bluetooth is called **Piconet**. **PAN**s and **WPAN**s usually extend only a few meters and are therefore not suitable for connecting devices in separate rooms or even buildings.

In the context of the **Internet of Things** (**IoT**), **WPAN**s are used to communicate control and monitor applications with low data rates. Protocols such as Insteon, Z-Wave, and ZigBee were explicitly designed for smart homes and home automation.



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